

PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)
III B.Tech I Semester Regular Examinations, November - 2025
DATA WAREHOUSING & DATA MINING
(CSE)

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) Define Data Warehouse.	K1	CO1	2M
	b) List the characteristics of a Data Warehouse.	K1	CO1	2M
	c) Define data cleaning.	K1	CO2	2M
	d) What is data integration?	K1	CO2	2M
	e) Define classification.	K1	CO3	2M
	f) Define prediction and how it differs from classification.	K2	CO3	2M
	g) Define association rule mining.	K1	CO4	2M
	h) What are support and confidence?	K1	CO4	2M
	i) List the types of clustering methods.	K1	CO5	2M
	j) What is the difference between partitioning and hierarchical methods?	K2	CO5	2M
UNIT – I				
2.	a) Explain the architecture of a Data Warehouse with a neat diagram.	K2	CO1	5M
	b) Compare star schema, snowflake schema, and fact constellation schema.	K3	CO1	5M
OR				
3.	a) Evaluate the performance of Data Warehouse design techniques.	K2	CO1	5M
	b) Explain various OLAP operations with suitable examples.	K2	CO1	5M
UNIT – II				
4.	a) Explain the different steps involved in data cleaning.	K2	CO2	5M
	b) Discuss various data transformation techniques with examples.	K3	CO2	5M
OR				
5.	a) Discuss common issues faced during data integration and suggest remedies.	K3	CO2	5M
	b) Explain how preprocessing improves the performance of mining algorithms.	K2	CO2	5M
UNIT – III				
6.	a) Describe attribute selection measures used in decision tree algorithms.	K3	CO3	5M

	b)	A classifier on 200 test samples gives: True Positives = 70, True Negatives = 90, False Positives = 20, False Negatives = 20. Compute: Accuracy, Precision, Recall (Sensitivity), Specificity, F1-score.	K3	CO3	5M
OR					
7.	a)	Explain the process of building a decision tree with a simple example.	K2	CO3	5M
	b)	Compare decision tree and Bayesian classification approaches.	K3	CO3	5M
UNIT – IV					
8.	a)	Discuss the challenges involved in association rule mining.	K3	CO4	5M
	b)	Analyze the steps in generating strong association rules from large datasets.	K4	CO4	5M
OR					
9.	a)	Discuss the importance of association analysis in business decision-making.	K3	CO4	5M
	b)	Explain the problems encountered in mining multilevel association rules.	K4	CO4	5M
UNIT – V					
10.	a)	Explain the steps involved in K-means clustering with a suitable example.	K2	CO5	5M
	b)	Describe the agglomerative hierarchical clustering algorithm.	K2	CO5	5M
OR					
11.	a)	Explain the working of the DBSCAN clustering algorithm.	K2	CO5	5M
	b)	Compare K-means and K-medoids clustering methods.	K3	CO5	5M